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Dynamically Reconfigurable WSN Node Based on ISO/IEC/IEEE 21451 TEDS

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***Abstract*—Ubiquitous and plug-and-play are fundamental characteristics for the development of Wireless Sensor Networks (WSNs). The ISO/IEC/IEEE 21451 Family of Standards has recently awakened its interest in this development, making easier the introduction of both characteristics in the design of wireless sensor nodes. Dynamic reconfigurability is also an exciting topic that is gathering interest for the development of embedded systems and smart sensor nodes in the last few years. The use of dynamic reconfiguration increases the scalability and heterogeneity capabilities of the sensor network. However, the ISO/IEC/IEEE 21451 Standards does not handle dynamically reconfigurable nodes and previously proposed frameworks do not include this capability into novel wireless sensor nodes. In this work an ISO/IEC/IEEE 21451 Transducer Electronic Data Sheet (TEDS) architecture for the management of reconfigurable WSNs nodes is proposed. In order to test the node architecture and validate its capability, the proposed TEDS was implemented for an environmental sensor network.**

***Index Terms*— ISO/IEC/IEEE 21451-x, Reconfigurable Architectures, WSNs.**

I. Introduction

WIRELESS Sensor Networks (WSNs) have become an interesting research topic with implications in key technological areas such as telemedicine, automotive, environmental, domotics or military since their origin during the early 1980's into the Distributed Sensor Networks program at the Defense Advanced Research Projects Agency (DARPA) [1–3]. Their main goals include generating, transferring, analyzing and processing information from heterogeneous sensor nodes, using wireless networks to have signal processing, localization, or object and event tracking capabilities [4-5]. With the proliferation of sensor networks and sensor networks applications, the integration between heterogeneous sensors networks is currently a key challenge in this area [6]. Consequently, the use of international standards for the definition of interfaces and communications protocols in distributed instrumentation environments is becoming a relevant point, since they improve the integration between heterogeneous sensor nodes and ensure the interoperability between different sensor networks. Towards these goals, the IEEE and the National Institute for Standards and Technology (NIST) have pioneered the development of a set of standards known as ISO/IEC/IEEE 21451 (previously known as IEEE 1451.x) [7-11]. The ISO/IEC/IEEE 21451, is a family of Smart Transducer Interface Standards, these standards describe a set of open, common, network-independent communication interfaces for connecting transducers (sensors or actuators) to microprocessors, instrumentation systems, and control/field networks. Although the ISO/IEC/IEEE 21451 Standards specify the basis for the interoperability and scalability in heterogeneous sensor networks, many issues are not yet addressed and more research are required, given that WSN applications continuously demand more capabilities and performance of the network.

Normally, an embedded processor, a communication device and some interfaces, constrained to minimize power consumption, establish a WSN node architecture. However, due to the demand of new applications that require more computational power and more adaptability of the node [12], this approach has shifted in recent years. Adding flexibility and scalability into the node architecture, can improve the performance of the sensor network, consequently extensive work has been conducted recently to develop flexible designs [13-15]. Several issues limit the development of flexible architectures, for example, the complexity of the integration and handling of reconfigurable devices in the node. Reconfigurable hardware has been analyzed as an alternative for application-specific integrated circuit (ASIC) in WSN node. For fixed functionality, ASIC can deliver the best performance with optimal power efficiency over a reconfigurable architecture, but with limited flexibility [13]. Yet a reconfigurable architecture facilitates the scalability in broad applications compared with ASIC. Furthermore, flexibility is facilitated as they can be remotely reconfigured after deployment while simultaneously shortening the time-to-market by skipping the manufacture process during development.

In the literature, several studies have been carried out to give reconfiguration capabilities to WSN nodes. For example, a Field Programmable Gate Array (FPGA) is proposed in [13] for adding reconfigurable capability to the sensor, jointly with a ZigBee module for communicating the acquired data. There

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are others even more complex end-user proposals systems. In [16] the processing task is done using a FPGA that includes a digital function library to connect with the conditioning and filtering stages. The networking tasks are implemented on an ADUC841 microprocessor, requiring a hardware-level implementation of the protocols. In the RESENSE research project, a modular reconfigurable platform has been developed [14]. In this platform, a low cost ATMega microcontroller performs the processing task, and a reconfigurable HW (a CPLD device) is used to achieve communication abstraction through the entire modular platform. The UISI project presents a Universal Intelligent Sensor Interfaces inspired by the IEEE 1451.3 standard. By using a PSoC 1 (Programmable System on Chip) device to manage the different electronic interfaces of the sensors available in the market, this platform provides a flexible analog and digital front-end including conditioning and conversion functions [15].

Although these platforms have reconfigurable capabilities in order to manage different proprietary interfaces, making the integration of heterogeneous sensors easier, they are all based on proprietaries protocols in order to support sensor nodes reconfiguration capabilities. Currently, there are no standards for handling reconfigurable devices in WSN platforms. In order to standardize the incorporation and the management of reconfigurable nodes in WSNs, in this work we propose and implement a novel template based on the ISO/IEC/IEEE 21451 Standards Transducer Electronic Datasheet (TEDS). The proposed innovation exploits the ISO/IEC/IEEE 214510 Standard flexible structures for the inclusion of a new class of TEDS that, along with a programmable hardware register-based architecture, allows for managing the reconfiguration process in WSN nodes.

The goal of this proposal is to turn a sensor node into an intelligent sensor with a reconfiguring dynamic capability. To validate the concept, the proposed TEDS was implemented and tested in an environmental sensor network. For this test-bed, we chose a Programmable System on Chip (PSoC) architecture, as the reconfigurable hardware. The PSoC device integrates analog and digital programmable blocks, which are very useful for implementing embedded systems. The results show that the proposed TEDS significantly simplifies both, the management of the reconfiguration process, and the integration of new transducers, resulting in improved interoperability within the WSNs. This paper makes a twofold contribution. First, we propose a novel TEDS template for managing reconfigurable node architecture in WSN. Secondly, we implement and evaluate a WSN node based on the proposed TEDS using a PSoC device.

This article is organized as follows: after introducing the problem of the interoperability of sensor networks in Section I, Section II presents an overview of the ISO/IEC/IEEE 21451 Standards and the benefits obtained by following it in the design of smart WSN architectures. Section III presents the design of the proposed TEDS for reconfigurable node architecture based on the ISO/IEC/IEEE 214510 Standard, detailing its characteristics and looking for possible technologies to implement the WSN node. Subsequently, in Section IV, the implementation of a reconfigurable architecture for WSN node using a PSoC device is done. The results and discussions of several experiments conducted in the laboratory and in real applications are presented in Section V. Finally, the conclusions are provided in Section VI.

II. Smart Sensor and the ISO/IEC/IEEE 21451 Standards

The smart sensor concept has been introduced to simplify the interoperability and scalability of networked electronic equipment [6]. Smart sensors can be defined as sensors having data conversion, information processing, and communication abilities. Smart sensors are also defined by the ISO/IEC/IEEE 214510 Standard as sensors with limited resource and standardized physical interfaces to enable the communication with data networks. Generally, a smart sensor includes elements like a physical transducer, a processor/memory core and a network interface. In particular, the “plug and play” capability of a 21451-compliant transducer is the major difference between a smart sensor defined by the standard and a typical sensor. Fig. 1 shows the elements of a smart sensor as established by the ISO/IEC/IEEE 21451 Family of Standards [17].

The ISO/IEC/IEEE 21451 family of Standards defines a set of common interfaces, functionalities and commands for connecting sensors and actuators to microprocessor-based systems, instruments, and field networks in a network-independent fashion. The main goal of this family of standards is to develop networks and transducer interfaces vendor independent, so that transducers could be replaced and/or exchanged with minimum effort. Another desirable functionality is the inclusion of a general transducer data, control, timing, configuration and calibration model in order to avoid error-prone manual configuration steps. The TEDS concept is developed by the standard as an electronic datasheet attached to the transducer, allowing the real-time access to data and calibration details. Two major components are defined, the Network Capable Application Processor (NCAP) and the Transducer Interface Module (TIM). The NCAP represents a network node responsible for the application processing. Fig. 2 shows a general overview of the Standards family.

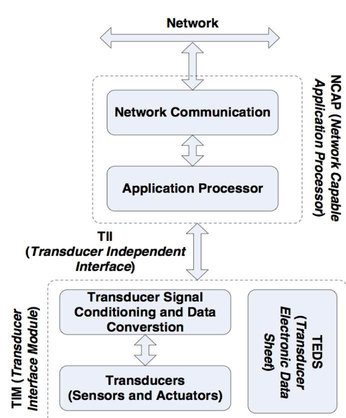

Fig. 1 Smart sensor architecture based on IEEE 1451 (Song et al, 2008 [16]).

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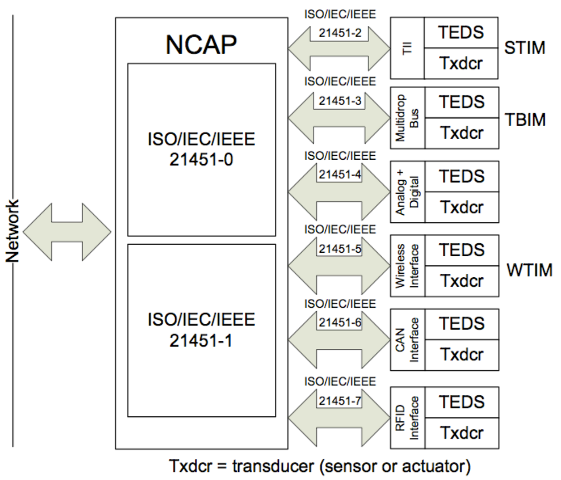

Fig. 2 General overview of IEEE 21451-x Standards family.

Several separate Standards are included in the ISO/IEC/IEEE 21451 family. ISO/IEC/IEEE 214510 defines a common set of commands –along with a communication protocol– to access transducers in the TIM, and it also defines the TEDS structure. The aim of ISO/IEC/IEEE 214510 is to encourage compatibility across the ISO/IEC/IEEE 21451-x family [7]. The ISO/IEC/IEEE 21451-1, currently under revision, defines a common object model for smart transducers along with interface specifications for a model component [8]. It also defines the NCAP in order to manage the transducers within the networks.

Different physical interfaces are possible between NCAP and TIMs, being this one of the most significant characteristics of the standard. For instance, ISO/IEC/IEEE 21451-2 defines serial interfaces for point-to-point communication between transducers to a NCAP [9] known as the Transducer Independent Interface (TII). The ISO/IEC/IEEE 21451-4 presents a mixed-mode communication protocol for analog transducers with analog and digital operation [10], and it defines a mechanism for adding self-identification technology to traditional analog sensors and actuators. ISO/IEC/IEEE 21451-5 defines wireless communication methods, including only 802.11 (Wi-Fi), 802.15.1 (Bluetooth) and 802.15.4 (ZigBee) as physical interfaces. However, other physical interfaces and wireless protocols could be incorporated [18]. Finally, other well-known interfaces such as CAN-Open or the radio frequency identification (RFID), also meet the proposed ISO/IEC/IEEE 21451-6 and ISO/IEC/IEEE 21451-7 Standards [11].

Since the development of the ISO/IEC/IEEE 21451-x family of standards, several research projects have been carried out [19-23]. For instance, an IEEE 1451.0 standard-based framework was proposed for Intelligent Transportation System applications. This framework follows the standard self-contained criteria and it takes the IEEE 1451.0 standard as an object-oriented model, simplifying this way the implementation of the framework [23].

It is important to emphasize that the complex level of the ISO/IEC/IEEE 21451-x Standards has troubled its implementation. However, the standard structure was designed to facilitate the implementation of WSN applications. The TEDS concept makes easier the configuration of the sensor node, ensuring the *plug-and-play* capability of the smart sensors. Additionally, the TEDS structure could be used for other kind of tasks such as recognizing signal patrons or managing reconfigurable nodes platform. Currently, the standard allows the implementation of new kinds of TEDS for proprietary solutions. In this work, we propose using TEDS template and its structure for the handling of reconfigurable nodes architectures.

III. Node Reconfigurability in WSN

Much work has been done for the development of more versatile and flexible wireless node architectures [24-27]; the run-time reconfiguration is often considered the best solution for these requirements. Since reconfiguration is usually made after nodes deployment, radio-reprogramming methods are often useful being the most common one [28]. However, the main disadvantage of this approach is that, regardless of the modification size, full application must be transmitted and downloaded taking over a considerable portion of the available network bandwidth. Another drawback is that the running program must be stopped at least briefly for reprogramming. Due to these disadvantages, the reprogramming process is rarely used. As an alternative approach for the reconfiguration process, we focus on the HW conditioning circuit, changing only the selected HW blocks trough the proposed TEDS.

Since the transducers on each sensor node are the interfaces between the physical environment and the WSN system sensor, nodes are required to measure different variables. In most cases, WSNs application involves analog measurement from transducers, including temperature, humidity, pressure, etc. These transducers usually require a conditioning circuit prior to data processing. Programmable mixed-signal (or custom CMOS) hardware with on-the-fly reconfigurability capability is the enabling technology used to implement this proposed TEDS. Compared to the traditional WSN nodes hardware, the hardware resources could be optimally used through the real-time reconfiguration. By reconfiguring the hardware structures, the same programmable mixed signal circuits can be reused for supporting different types of transducers different time instances. The proposed TEDS also have the potential to support new transducers that are not included in the original design, avoiding in this way installing new hardware for the addition of new transducers. This feature can greatly increase the flexibility and scalability of the system, while reducing its complexity, size and cost.

In order to implement the proposed TEDS, two major commercially available programmable hardware technologies of the analog domain, Field Programmable Analogue Arrays (FPAA) and Programmable Mixed Signal System on Chip technology (e.g. PSoC) have been considered. The new FPGA technology incorporating mixed-signal capability has also been considered but its analog capability is limited and not

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adequate for typical WSN applications. Both FPAA and PSoC technologies are capable of providing all the typical components for supporting different analog sensors, but compared to FPAA, PSoC possesses features that are ideally suitable for WSN applications. In this work, PSoC technology was chosen due to the following characteristics:

1) Lower power consumption and lower cost than FPAA.
2) PSoC has a built-in CPU subsystem with SRAM, EEPROM and flash memory in addition to configurable analog blocks while FPAA requires an external microcontroller.
3) PSoC contains programmable digital blocks that can be reconfigured into different digital and communication peripheral functions such as timers, PWMs, I2C, SPI, UART, etc.
4) Digital processing capabilities and interfaces are integrated.

In addition, in the PSoC devices programmable blocks are mapped into registers, so by setting values into the registers it is possible to change the hardware connections between the blocks. This is also possible for a FPAA, but considering the above listed characteristics, a PSoC device was preferred, and therefore, selected.

A. Data structure of the reconfiguration parameters

Once the device to be used is selected, it is necessary to define the handling of the reconfiguration process. In a WSN node, the reconfiguration process depends of several factors, such as the definition of an appropriate data structures and functional commands, in relation to the limited resources featured in wireless networks. Consequently, an appropriate data structure is required to store the node's reconfiguration parameters. The use of an international standard can simplify the reconfiguration process and ensures the integration of new transducer interfaces improving the interoperability and scalability of the wireless sensor nodes. In this work, the ISO/IEC/IEEE 214510 standard TEDS is used for handling reconfigurable nodes. It is important to note that besides being used as a conditioning hardware identifier, the proposed TEDS can be used to perform the reconfiguration process in the sensor node. Usually, TEDS concept is defined as an electronic datasheet that allows access to data calibration and manufacturer-defined specifications. However, TEDS structure fulfils the requirement to store the reconfigurable conditioning circuit connexions between the blocks. Two main characteristics are desirable in this reconfigurable TEDS: (1) a hardware-level self-identification mechanism in the node and (2) a parameter to perform a hardware-level reconfiguration process. Therefore, the new proposed TEDS can be interpreted as an electronic version of the conditioning module netlist. Fig. 3 shows the concept of the reconfigurable TEDS. This research is focused on the analog hardware blocks provided by the PSoC 5 LP device; this device provides several analog programmable blocks.

PSoC devices can be programmed trough the PSoC Creator software IDE by Cypress [29]. This software contains a graphical design editor for producing hardware/software co-design projects. The PSoC Creator consists of two basic building blocks. A software that allows the user to select, configure and connect existing circuits on the chip and, the components, which are the equivalent peripherals on MCUs.

However, PSoC Creator IDE cannot handle runtime configuration blocks. In order to achieve this, a low-level configuration of the blocks connection must be performed. PSoC functional blocks and routing configurations paths are mapped into specific registers, so by setting values in these registers, it is possible to configure and connect the blocks between them. The number of registers to be configured depends on the block, but a common subdivision of registers includes Configuration Registers (CR), Routing Registers (SW) and Power Mode Registers (PM). GPIO ports also have their configuration registers allowing the connection between the blocks and to other components external to the chip. This great flexibility enables us to configure a PSoC device.

To match the proposed TEDS with the PSoC architecture, a mapping process is required. Table I shows the required bytes needed to define the PSoC blocks. The mapping process is applied in order to describe the hardware blocks and their connections. The results of this mapping are structured into the fields of the proposed TEDS according to the ISO/IEC/IEEE 214510 format, as explained in the following section. The TEDS contains several fields that describe the different analog blocks, such as Op-amps, DAC, ADC, SW-amps, and the I/O ports.

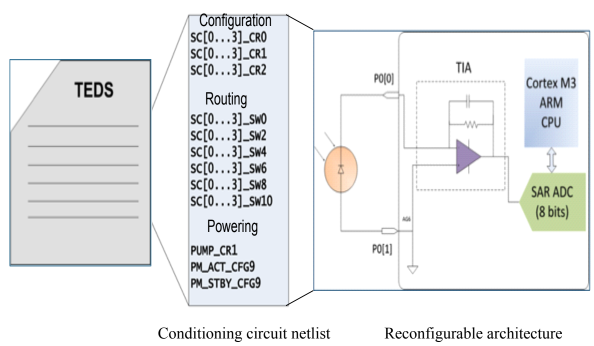

Fig. 3 Proposed reconfigurable TEDS of a conditioning circuit.

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TABLE I
DATA STRUCTURE REQUIRED TO DEFINE PSOC BLOCKS

Block	Size (Bytes)	Description
SC/CT	32	Amplifier kind (PGA, Inv-PGA, TIA), gain, power level, routing
Op-amp	20	Functional mode, power level
SAR ADC	16	Resolution, routing
Delta Sigma ADC	8	Resolution, sample rate, routing
DAC	20	Operation range, conversion value
GPIO	5	Routing information
Others	6	Reserved

Table II shows in more detail the Op-amp field of PSoC architecture. This field contains all the information required to configure an operational amplifier circuit. It takes only five octets to complete the configuration of a single Op-amp circuit. The ON/OFF field allows for enabling or disabling the Op-amp block, the Mode field defines the configuration of the Op-amp (Inverter, Not inverter or Follower); the Power field configures the power level of the Op-amp output. The last two fields indicate the way negative and positive terminals are connected (either to an analog bus, or to internal reference).

TABLE II
OP-AMP FIELD DATA STRUCTURE

Field	Value	Description
On/Off	--	Indicate if the Op-amp is enabled or not
Mode	0x00 Naked Op-amp 0x01 Follower 0x02 Inverter 0x03 Not inverter	This field indicates the configuration mode of the Op-amp
Power	0x00 Minimal 0x01 Lower 0x02 Medium 0x03 Max	This field indicates the power level of the Op-amp
Vn	0x00 Not connected 0x01 Analog global 0x02 Analog global	This field indicates where the negative terminal of the Op-amp is connected
Vp	0x00 Not connected 0x09 VREV 0x01 Analog global1 0x02 Analog global2 ……. 0x07 Analog global7	This field indicates where the positive terminal of the Op-amp is connected

IV. RECONFIGURABLE SMART SENSOR NETWORK IMPLEMENTATION

Two entities, defined by the standard ~~a~~ Wireless Transducer Interface Module (WTIM) and the NCAP, are required to integrate the ISO/IEC/IEEE 21451 Standards with the proposed reconfigurable WSN node architecture.

A. WTIM Implementation

The WTIM is integrated into a smart sensor node (Fig. 4) while NCAP is implemented in the network sink node. The implementation of a reconfigurable WTIM requires the use of a hardware device (with analog or digital reconfigurable blocks for signal conditioning), a microcontroller (for performing data acquisition and for processing tasks) and a communication module (for transmitting data from the transducers to the NCAP). In this work, a PSoC architecture is chosen as the reconfigurable device while a TelosB mote was selected as the communication module.

The PSoC 5 LP devices have a Cortex-M3 CPU, a digital system that includes Universal Digital Blocks (UDBs), specific function peripherals (including USB, and CAN). Also an analog system that includes Delta Sigma converters, analog to digital SAR converters and Switched Capacitor - Continuous Time (SC/CT) blocks. A simplified block diagram of PSoC 5 architecture is presented in Fig. 5.

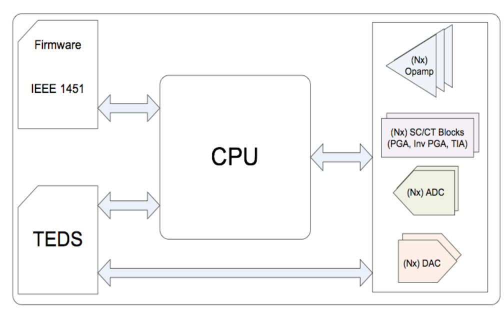

Fig. 4 Proposed WTIM reconfigurable node architecture.

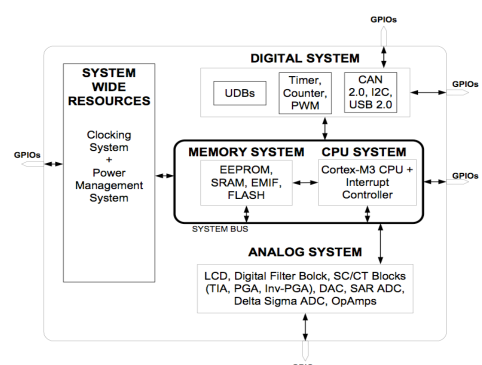

Fig. 5 Top-level architecture of PSoC 5 LP device. Cypress Semiconductor [29]

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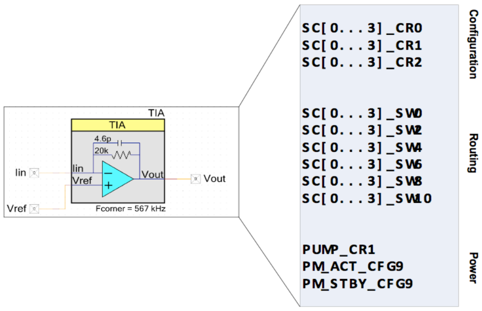

Fig. 6 Reconfiguring of PSoC 5LP registers to establish a TIA amplifier

The Fig. 6 shows, the registers to be configured in order to establish a TIA amplifier with the PSoC 5 blocks. Table III presents all the analog PSoC 5 LP blocks mapped in the proposed reconfigurable TEDS according to the PSoC data structure defined in the above section and the ISO/IEC/IEEE 214510 standard. The blocks setup include four SC/CT blocks, a general-purpose block constructed by an amplifier with arrays of components, two SAR ADC blocks, a Delta Sigma ADC, four DAC blocks and the GPIO ports of the chip. The reconfigurable TEDS can be sent from the NCAP to the WTIM in order to implement a new conditioning circuit on the node. Pre-configured conditioning circuits can also be stored in the WTIM as default configurations. Based on the structure defined by the standard, we were able to map all the analog hardware resources of the PSoC 5 LP into the proposed TEDS by using only 107 bytes.

To complete the wireless node, we selected a TelosB based platform, the CM3300 network interface, which presents a wide range coverage. This platform is based on both, the MSP430 ultra-low-power microcontroller and the CC2420, a low-power IEEE 802.15.4 radio chip. Several research groups use this platform to develop new protocols and it became a *de facto* standard in WSNs. The communication between the CM3300 mote and the PSoC device is established through a serial port and a rechargeable battery powers all the WTIM. In order to reduce the WTIM power consumption, a wake-up signal pin is integrated between the CM3300 and the PSoC. Since the CM3300 activates the PSoC device only when a measurement is performed, the WTIM power consumption is almost the same as the TelosB platform.

B. NCAP design

The NCAP is a standard entity that allows remote accessing to the smart sensor node by another network such as the Internet. The NCAP was implemented in a PC connected to the sink node of the network. All the data received by the sink from the network are sent to the NCAP for storage and display purposes. The sink node is also responsible for sending to the nodes commands generated by the NCAP, such as reading data commands, TEDS commands and reconfiguration commands. The NCAP services are implemented in C#, and a database server was developed in order to storage the sensors measurements.

Once started, the NCAP reads the initial configurations stored in the database (number of nodes, data from sensors), and then it is ready to interact with the sensor network. As soon as a node is connected to the network, a MetaTEDS and network ID are generated and broadcasted to the NCAP. Subsequently, the NCAP receives these messages, stores the data and enrolls the node as a new node. In the database, two kinds of tables are defined. The first contains all the data related to the WTIM (ID and TEDS) while the second contains data from the sensor nodes, the channel ID and time-stamp of the measurements. Furthermore, a user-friendly GUI was implemented. This interface allows sending ISO/IEC/IEEE 21451commands and it is able to process the data sent by the nodes to the NCAP.

Two different kinds of reconfiguration processes are developed. The first one, based on an algorithm in the firmware of WTIM, begins the reconfiguration process triggered by a local event. A predefined library containing different sensors TEDS, which could be mounted on the WSN node, is preloaded. In the second one, a request from the NCAP begins the reconfiguration process sending the sensor TEDS through the network. For instance, when the application involves measurements of temperature, a set of different temperature transducers TEDS are preloaded in the node. The TEDS library contains all the data required to establish one possible conditioning circuit. This reconfiguration is performed by the proposed AddGenericSensor command. Once the WTIM receives the TEDS, it assembles the hardware on the reconfigurable device according to the TEDS.

In order to support the reconfiguration process, the NCAP entity should be able to handle two new commands, the AddSensor and AddGenericSensor commands, which are both added to the Standard. The Add generic sensor GUI (Fig.7) allows the user to define new conditioning circuits. This GUI contains a generic list of analog blocks that allows for setting the configuration of parameters such as gain or input reference. In addition, it is possible to set the routing path of the circuit.

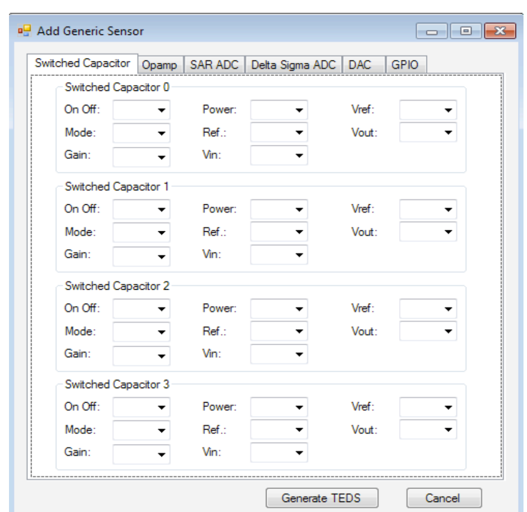

Fig. 7 Add Generic Sensor GUI.

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TABLE III
PROPOSED TEDS FIELDS STRUCTURED ACCORDING TO THE ISO/IEC/IEEE 214510 STANDARD AND THE PSOC DATA STRUCTURE

Field	Name	Description	Type	Octets
-		Length	UInt32	4
0 -2		Reserved	-	-
3	SC0	Switched Capacitor/ Continuous Time block 0	UInt8	8
4	SC1	Switched Capacitor/ Continuous Time block 1	UInt8	8
5	SC2	Switched Capacitor/ Continuous Time block 2	UInt8	8
6	SC3	Switched Capacitor/ Continuous Time block 3	UInt8	8
7	Op-amp0	Op-amp block0	UInt8	5
8	Op-amp1	Op-amp block1	UInt8	5
9	Op-amp2	Op-amp block2	UInt8	5
10	Op-amp3	Op-amp block3	UInt8	5
11	SAR0	SAR ADC0	UInt8	8
12	SAR1	SAR ADC1	UInt8	8
13	DSM0	Delta Sigma ADC 0	UInt8	8
14	DAC0	DAC block 0	UInt8	5
15	DAC1	DAC block1	UInt8	5
16	DAC2	DAC block 2	UInt8	5
17	DAC3	DAC block 3	UInt8	5
18	Port0	Port 0 Information	UInt8	1
19	Port2	Port 2 Information	UInt8	1
20	Port3	Port 3 Information	UInt8	1
21	Port4	Port 4 Information	UInt8	1
22	Port6	Port 6 Information	UInt8	1
--		Checksum	UInt16	2

After completing this setting, the user can generate the new TEDS by pressing the Generate TEDS function, which is then saved in the NCAP database. Finally the netlist gets sent wirelessly to the corresponding WTIM for its implementation.

By using an ID Sensor seller as parameter, the AddSensor command requests from the TEDS libraries those preloaded hardware details related to conditioning circuit (Fig. 8). In this way, the firmware reads the reconfiguration TEDS in order to assemble the hardware and leave the sensors ready to begin taking data. After this command, the smart sensor exits the SLEEP state.

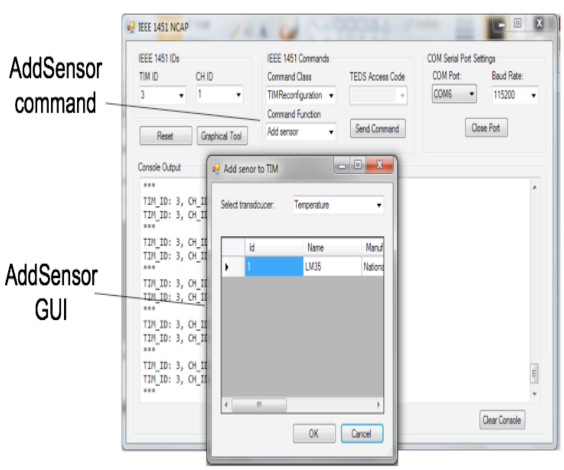

Fig. 8 NCAP GUI interface. On the top, the add sensor panel.

Additionally, in order to dismantle the WTIM, a new RemoveSensor function was included. With this function, it is possible to remove sensors conditioning hardware that are not necessary, enabling the efficient use of the reconfigurable resources. The new commands require a communication sequence, i.e a handshaking procedure. These procedures ensure the maintenance of the TEDS NCAP database updated after the reconfiguration. Fig. 9 shows the implemented communication sequence.

V. RESULTS AND DISCUSSION

This section presents some experimental results for managing a reconfiguration process as applied to different kinds of transducers used to monitoring environmental variables. Two adaptive reconfiguration processes are implemented in this work. The first one requires a request from the NCAP to generate the HW reconfiguration in the WTIM. The second reconfiguration technique is based on an algorithm implemented in the WTIM firmware. In the first technique, once the TEDS arrive in the WTIM, the firmware interprets the proposed TEDS and implements the HW reconfiguration. This allows the capability for implementing new conditioning circuits for new transducers that are not initially included in the NCAP database TEDS. Fig. 10 illustrates the simplified state machine implemented in order to accomplish a reconfiguration process, once the WTIM has received the reconfiguration NCAP command.

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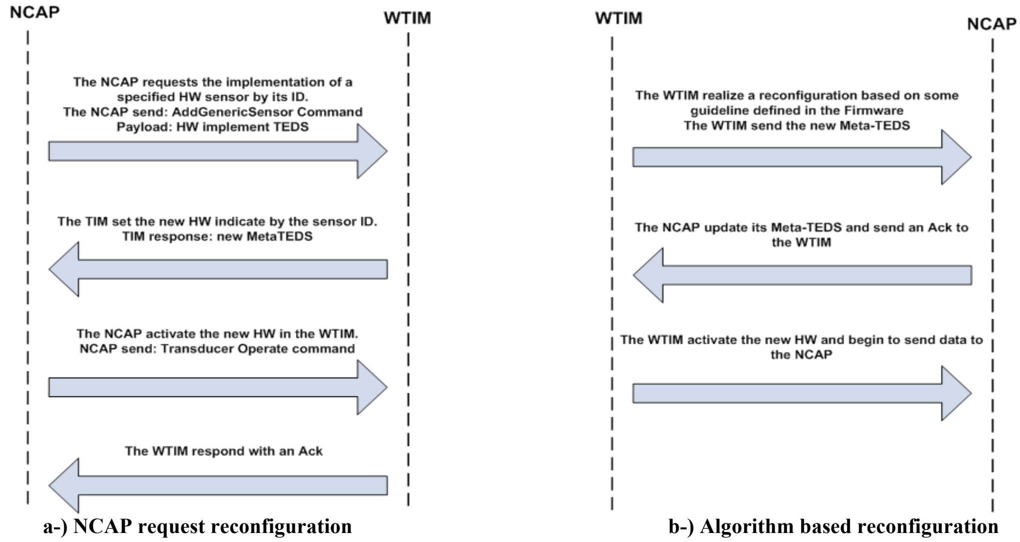

Fig. 9 Communication sequence necessary for updating the TEDS. a- NCAP request reconfiguration. b- A smart algorithm based.

The WTIM is capable of accomplishing the reconfiguration process in millisecond intervals, after which the firmware proceeds normally.

In the second technique, a WTIM algorithm based produces the reconfiguration process triggered by an event. Fig. 11 shows the state machine representing the WTIM's behavior in response to changes in the measured variables. In this experiment, as the temperature is below 308.15 Kelvin (35-Celsius) degrees, the WTIM registers values of the temperature and humidity with just 20 minutes sampling time, intending to save power battery. But when the temperature is above 308.15 Kelvin (35-Celsius) degrees, the sampling time decreases to 5 minutes, anticipating that an increase in the temperature might indicate the beginning of an overheated state. As the temperature reaches 310.15 Kelvin (37-Celcius) degrees, the WTIM reconfigure its HW, establishing the conditioning circuit (which is not defined by default) necessary to measure carbon monoxide (CO), since it might indicate the beginning of a combustion phenomenon. The whole process is included by the modification of several HW blocks by WTIM, the updating of the reconfigurable TED and the Meta-TEDS and, the sending of the updated TEDS to the NCAP. This last step updates the TEDS database. The capability for establishing the HW required for CO measurement only as needed, allows for saving power battery, which is an important feature of the proposed architecture. Notice that this kind of transducers requires a 60 mA current and 5 minutes long heating cycle, before taking a measurement. Fig. 12 shows the results of some CO measurements during reconfiguration testing related to several ranges of temperature.

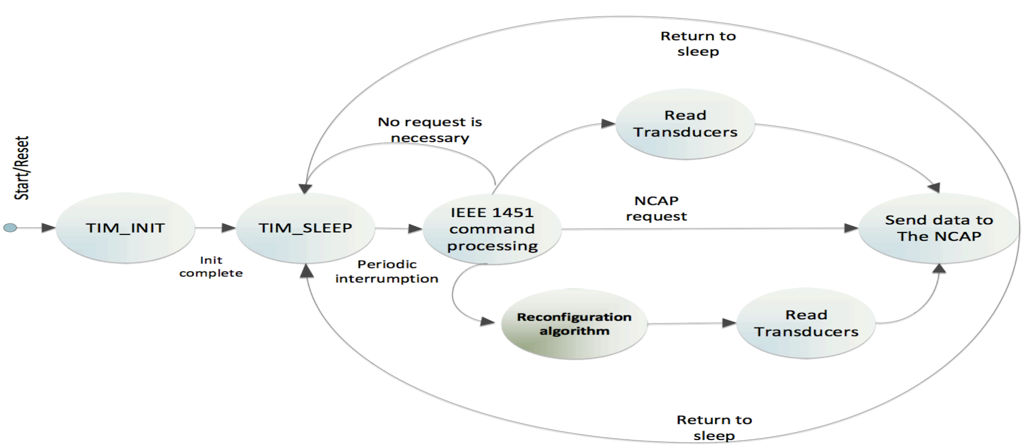

Fig. 10 WTIM state machine of the reconfiguration process by NCAP request.

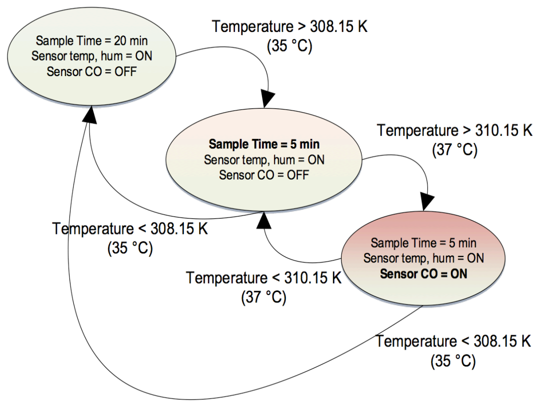

Fig. 11 WTIM state machine of the reconfiguration process activated by an event.

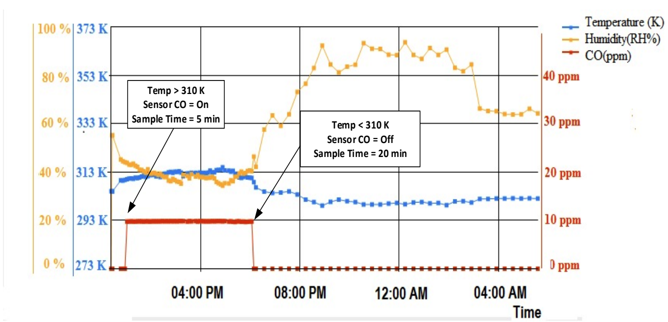

Fig. 12 Temperature, humidity and CO temporal signals. The graph shows the assembling and dismantling instants of the CO sensor triggered by a temperature threshold.

Fig. 13 shows the state machine of another experiment performed in order to highlight the advantages of a reconfiguration. This state machine shows events triggered by the node battery level. The node modifies its sampling time when the battery power level goes down to 90%. When the battery goes down to 85% the node dismantles the CO transducer conditioning circuit. Notice that the CO transducer power consumption is about the 60% of total node power consumption. Finally, when the battery power level goes down to 80%, the node dismantles the temperature conditioning circuit, turning the node into a simple relay node.

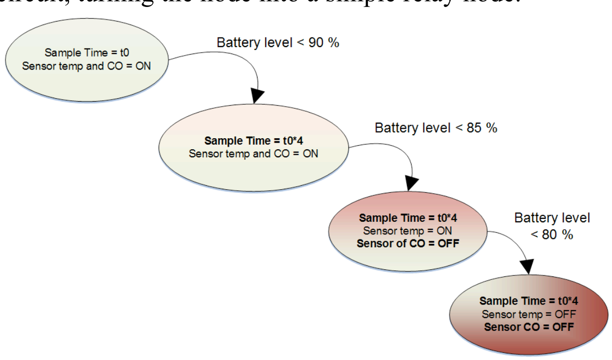

Fig. 13 WTIM state machine designed for saving battery power.

Finally, for the purpose of validating the performance of the proposed reconfigurable architecture, some circuit implementations were conducted. Table IV presents some circuit implementations and their results in terms of reconfiguration time and current consumption are presented. It is important to notice that, different conditioning circuits can be reconfigured in shorts periods of time.

These results shows that, reconfiguration can be performed *on-the-fly* either based on an algorithm or by a NCAP request. Consequently, high power conditioning circuits are established only when necessary, providing a mechanism for saving energy.

TABLE IV
MEASUREMENTS OF DIFFERENT CIRCUIT TOPOLOGIES IMPLEMENTATION

Functional Stage Topology	Configuration time (us)	Total current consumption (mA)
Sleep mode	-	0.002
Active, no circuit	-	19
Unit gain buffer + ADC	95	23.8
Two stage non inverting amplifiers + ADC	120	24

VI. CONCLUSION

The development of WSNs has experienced an important growth in recent years. However, their uses are limited by their dependency on proprietary interfaces and protocols. The combination of reconfiguration capability with the ISO/IEC/IEEE 21451 Family of Standards has the potential for generating a highly reconfigurable sensor node technology, ideal for wireless sensor networks nodes.

This work proposes a new TEDS template in order standardized the reconfiguration process in WSNs nodes. The implementation of this new TEDS into a reconfigurable adaptive wireless sensor node is presented. Application of this proposed architecture to an environmental sensor network shows that the use of the new TEDS increases sensor nodes flexibility while its implementation does not affect the network performance in issues related to power consumption and bandwidth requirement.

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